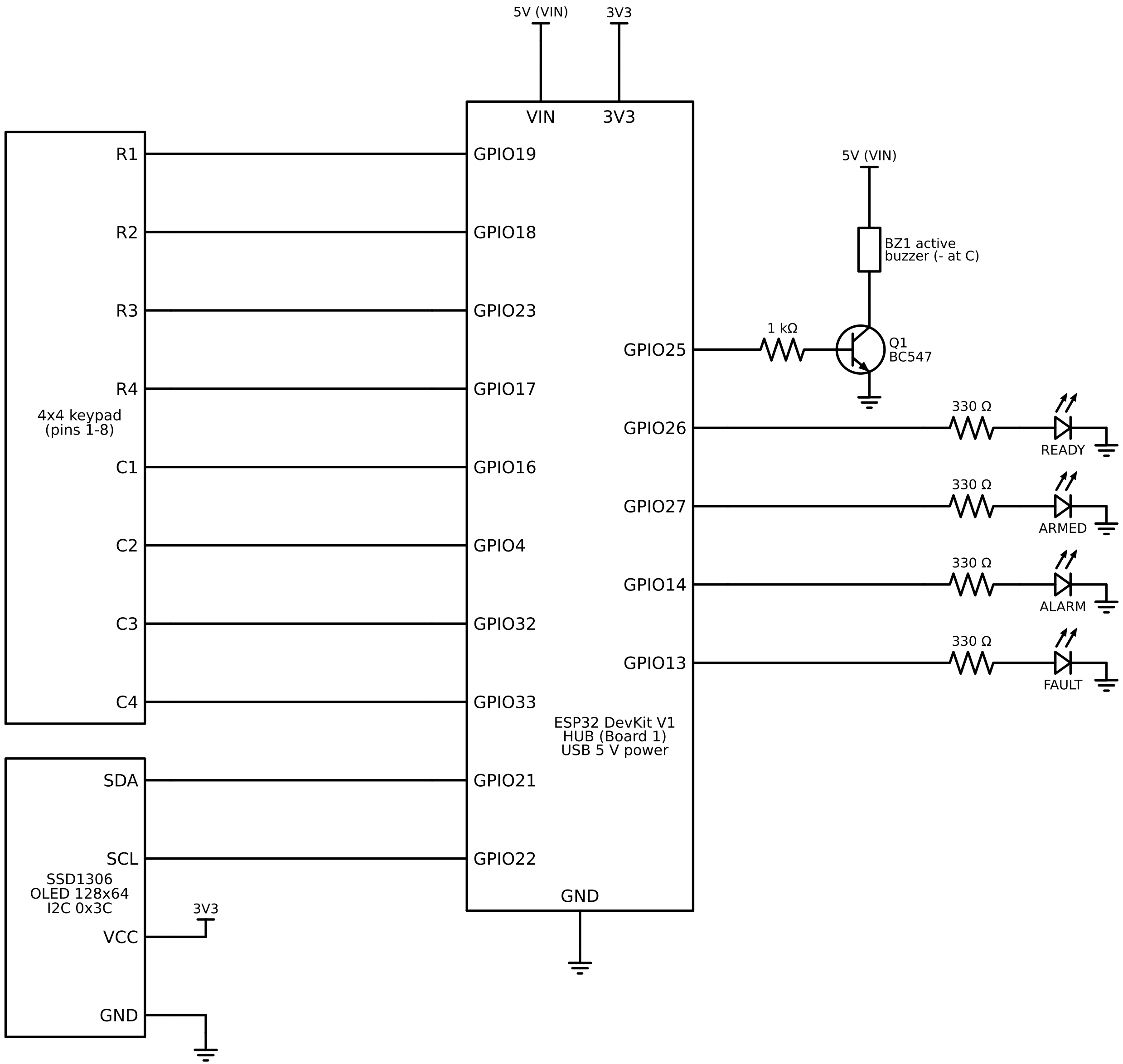

HomeGuard - HUB schematic (25ES604 ESD mini-project) rev 1 2026-11-05

Keypad pin 1-4 = rows R1-R4, pin 5-8 = columns C1-C4 (Keypad library pulls rows/cols internally). All GND symbols are the common ESP32 GND.

Siren: Ib = (3.3-0.7)/1k = 2.6 mA, BC547 saturated, Vce(sat) <= 0.2 V. LEDs: (3.3-2.0)/330 = ~4 mA each. Buzzer runs from 5 V VIN.